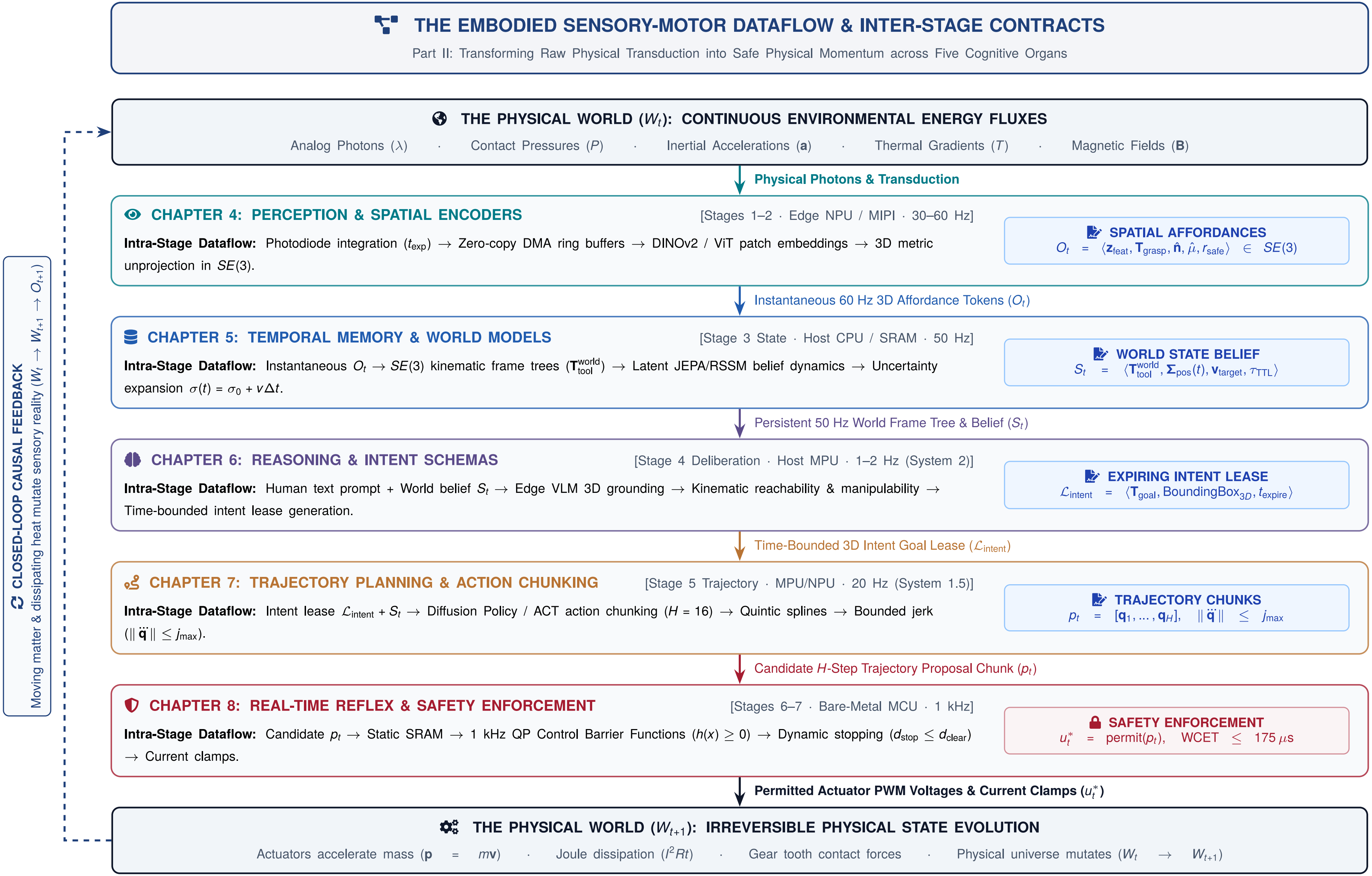

THE EMBODIED SENSORY-MOTOR DATAFLOW & INTER-STAGE CONTRACTS
Part II: Transforming Raw Physical Transduction into Safe Physical Momentum across Five Cognitive Organs
THE PHYSICAL WORLD (W_t): CONTINUOUS ENVIRONMENTAL ENERGY FLUXES
Analog Photons (λ) · Contact Pressures (P) · Inertial Accelerations ($\mathbf{a}$) · Thermal Gradients (T) · Magnetic Fields ($\mathbf{B}$)
Physical Photons & Transduction
CHAPTER 4: PERCEPTION & SPATIAL ENCODERS
[Stages 1–2 · Edge NPU / MIPI · 30–60 Hz]
Intra-Stage Dataflow: Photodiode integration (t_{exp}) → Zero-copy DMA ring buffers → DINOv2 / ViT patch embeddings → 3D metric unprojection in $SE(3)$.
SPATIAL AFFORDANCES
$O_t = \langle \mathbf{z}_{\text{feat}}, \mathbf{T}_{\text{grasp}}, \hat{\mathbf{n}}, \hat{\mu}, r_{\text{safe}} \rangle \in SE(3)$
Instantaneous 60 Hz 3D Affordance Tokens (O_t)
CHAPTER 5: TEMPORAL MEMORY & WORLD MODELS
[Stage 3 State · Host CPU / SRAM · 50 Hz]
Intra-Stage Dataflow: Instantaneous $O_t \to SE(3)$ kinematic frame trees ($\mathbf{T}_{\text{tool}}^{\text{world}}$) → Latent JEPA/RSSM belief dynamics → Uncertainty expansion $\sigma(t) = \sigma_0 + v\Delta t$.
WORLD STATE BELIEF
$S_t = \langle \mathbf{T}_{\text{tool}}^{\text{world}}, \mathbf{\Sigma}_{\text{pos}}(t), \mathbf{v}_{\text{target}}, \tau_{\text{TTL}} \rangle$
Persistent 50 Hz World Frame Tree & Belief (S_t)
CHAPTER 6: REASONING & INTENT SCHEMAS
[Stage 4 Deliberation · Host MPU · 1–2 Hz (System 2)]
Intra-Stage Dataflow: Human text prompt + World belief S_t → Edge VLM 3D grounding → Kinematic reachability & manipulability → Time-bounded intent lease generation.
EXPIRING INTENT LEASE
$\mathcal{L}_{\text{intent}} = \langle \mathbf{T}_{\text{goal}}, \text{BoundingBox}_{3D}, t_{\text{expire}} \rangle$
Time-Bounded 3D Intent Goal Lease ($\mathcal{L}_{\text{intent}}$)
CHAPTER 7: TRAJECTORY PLANNING & ACTION CHUNKING
[Stage 5 Trajectory · MPU/NPU · 20 Hz (System 1.5)]
Intra-Stage Dataflow: Intent lease $\mathcal{L}_{\text{intent}} + S_t$ → Diffusion Policy / ACT action chunking ($H = 16$) → Quintic splines → Bounded jerk ($\|\dddot{\mathbf{q}}\| \leq j_{\max}$).
TRAJECTORY CHUNKS
$p_t = [\mathbf{q}_1, \dots, \mathbf{q}_H], \quad \|\dddot{\mathbf{q}}\| \leq j_{\max}$
Candidate H-Step Trajectory Proposal Chunk (p_t)
CHAPTER 8: REAL-TIME REFLEX & SAFETY ENFORCEMENT
[Stages 6–7 · Bare-Metal MCU · 1 kHz]
Intra-Stage Dataflow: Candidate p_t → Static SRAM → 1 kHz QP Control Barrier Functions ($h(x) \geq 0$) → Dynamic stopping ($d_{\text{stop}} \leq d_{\text{clear}}$) → Current clamps.
SAFETY ENFORCEMENT
$u_t^* = \text{permit}(p_t), \quad \text{WCET} \leq 175\ \mu\text{s}$
Permitted Actuator PWM Voltages & Current Clamps (u_t^*)
THE PHYSICAL WORLD (W_{t+1}): IRREVERSIBLE PHYSICAL STATE EVOLUTION
Actuators accelerate mass ($\mathbf{p} = m\mathbf{v}$) · Joule dissipation (I^2Rt) · Gear tooth contact forces · Physical universe mutates ($W_t \to W_{t+1}$)
CLOSED-LOOP CAUSAL FEEDBACK
Moving matter & dissipating heat mutate sensory reality ($W_t \to W_{t+1} \to O_{t+1}$)